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SUPERSEDING
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MILITARY SPECIFICATION

NETWORK E CIRCUIT BOARD: 8623290

1. SCOPE

1.1 This specification covers one type of Network E Circuit Board, part of the Test Set, Computer Logic Unit, AN/GSM-70.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel; Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-46357	Computer, Gun Direction: M18: Manufacturing Processes and Inspection of

STANDARDS

Military

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U. S. Army Munitions Command

F8623290	Network E Assembly
F8623291	Schematic, Electrical, Network E
B8623458	Lamp Cartridges

FSC 1220

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PACKAGING DATA SHEET

M14002 Data Sheet, Master Packaging

(Copies of specifications, standards, drawings, and packaging data sheets, required by the suppliers in connection with specific procurement functions, should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 The Network E Circuit Board shall be manufactured in accordance with Drawing F8623290 and drawings pertaining thereto.

3.2 General specifications.- The circuit board shall meet the following requirements of MIL-F-13926, and MIL-C-46357.

3.2.1 MIL-F-13926

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Workmanship

3.2.2 MIL-C-46357

a. Etched circuit assembly
b. Marking
c. Handling

3.3 Environmental service conditions.

3.3.1 Vibration.- The circuit board shall be subjected to a simple harmonic motion having an amplitude of 0.015 inch (0.030 inch total excursion). The frequency shall be varied uniformly from 10 to 55 cycles per second (cps) and return to 10 cycles. This cycle shall be traversed in 1 minute ± 15 seconds. The motion shall be applied to the circuit board for a period of 10 minutes in each direction of the 3 mutually perpendicular axes (total of 30 minutes), relative to its normal operational position. Subsequent to vibration, the circuit board shall not be damaged or adversely affected and shall meet the requirements of 3.4.

3.3.2 Shock.- The circuit board shall withstand 3 shock impulses in the vertical plane relative to its normal mounting position. The acceleration of the shock impulses shall be 5 gravity units (G's). Pulse duration of each impulse shall be 11 ± 1 milliseconds. Subsequent to shock the item shall not have been damaged or adversely affected and shall meet the requirements of 3.4.

3.3.3 Temperatures.

3.3.3.1 Operating.- The circuit board shall meet the requirements of 3.5 while exposed to temperatures of -40°F and +135°F.

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3.3.3.2 Storage.- The circuit board shall meet the requirements of 3.4 at a temperature of between 60°F and 90°F after having been exposed to each temperature of -80°F and +160°F.

3.3.4 Humidity.

3.3.4.1 Operating.- The circuit board shall meet the requirements of 3.5 (high temperature conditions) while exposed to an atmosphere with a relative humidity (RH) up to 97% at a temperature of 135°F.

3.3.4.2 Non-operating.- The circuit board shall meet the requirements of 3.4 after exposure to (and when thoroughly dried) an atmosphere with a RH of 100 percent including condensation, for a period of 4 hours.

3.3.5 Fungus resistance.- The circuit board shall show no evidence of damage and shall meet the requirements of 3.4, after being inoculated with the composite spore suspension specified in Specification MIL-F-13926 and incubated at an ambient temperature of between 77°F and 84°F with a RH of 97 ± 2%.

3.3.6 Salt spray.- The circuit board shall show no evidence of physical damage and meet the requirements of 3.4 immediately after exposure to a fine mist of a 5% salt solution concentration. The solution shall have a specific gravity of from 1.0268 to 1.0413 and a pH between 6.5 and 7.2 at a temperature between 93° and 97°F.

3.4 Electrical performance.- The circuitry shall be in accordance with Electrical Schematic, Drawing F8623291. The symbols referenced in subsequent paragraphs pertain to this schematic and are used therein.

3.4.1 Error indicator.

3.4.1.1 True state - static conditions - (output loaded).- With a true input (minus 6 plus or minus 0.18 volts) and the load of figure 1 at the output, the neon load shall flash at a visible rate.

3.4.1.2 False state - static conditions - (output loaded).- With a false input (minus 0.15 plus or minus 0.15 volt) and the load of figure 1 at the output, the following conditions shall exist within the indicator:

(a) The base voltage shall be between minus 2.1 and minus 3.0 volts.
(b) The neon bulb at the load shall not be lighted.

3.4.2 Minus 18 volt driver.

3.4.2.1 True input - (output loaded).- With the input and load circuits of figure 2 connected to the proper pins, a true input shall effect:

(a) Plus 2.8 plus or minus 0.3 volts at the base of the input transistor.
(b) Plus 3 plus or minus 0.5 volts at the collector of the input transistor.

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(c) Zero plus or minus 0.5 volt at the emitter of the output transistor.

3.4.2.2 False input - (output loaded).- With a false input and the load of figure 2, the circuit parameters shall be:

(a) Plus 5.5 ± 1 volt at the base of the input transistor.
(b) Minus 18 ± 0.8 volts at the collector of the input transistor.
(c) Between minus 16.8, minus 18.5 volts at the emitter of the output transistor.

3.4.3 Oscillator (10 KC).

(a) The oscillator output shall meet the requirements of figure 3.
(b) The diode output (CR16), when grounded, shall cut off the output wave signal.

3.4.4 Phase detecting network.- With the loads of figure 4d applied, the following conditions shall apply to the circuit.

3.4.4.1 Phase one operation.- With the three phase input voltage applied to the input delta as shown in figure 4a, the following conditions shall exist within the network.

(a) the ac voltage E1 (refer to figure 4c) shall be no greater than 30 volts rms.
(b) The voltage E2 (figure 4c) shall be no less than 155 volts rms.
(c) The output voltage developed across the diode rectifier shall be no less than 130 volts dc.

3.4.4.2 Phase two operation.- With the input voltage sequencing changed as per figure 4b, the following conditions shall exist within the network:

(a) The voltage E1 shall be no less than 155 volts rms.
(b) The voltage E2 shall be no greater than 30 volts rms.
(c) The dc output voltage shall not exceed 15 volts dc.

3.4.5 Regular driver.

3.4.5.1 True state - static condition - (output loaded).- With a true condition (minus 6 plus or minue 0.18 volts) applied to the input gate of figure 5a, and the load of figure 5a, and the load of figure 5b at the output of the driver, the following conditions shall exist within a driver:

(a) The voltage at the collector of the input transistor shall be minus 1.6 plus or minus 0.1 volt.
(b) The voltage at the base of the output transistor shall be no less than 0.1 volt nor greater than 0.5 volt.
(c) With the load removed, the output transistor leakage current shall not exceed 15 microamps.

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3.4.5.2 False state - static condition - (output loaded).- With a false condition (minus 0.15 plus or minus 0.15 volt) applied to the input gate, and the output loaded, the following conditions shall exist within a driver:

(a) The voltage at the base of the output transistor shall be minus 0.52 plus or minus 0.23 volt.
(b) The voltage across the zener diode shall be 6.2 plus or minus 0.31 volts.
(c) The output transistor collector voltage shall be no less negative than minus 0.09 volt and no more negative than minus 0.48 volt.

3.4.6 Write switch conversion amplifier.- The following conditions shall exist for each of the 12 write switch conversion amplifiers.

3.4.6.1 True state - (output loaded).- With the load of figure 6b connected to the output of the amplifier, and a true signal (ref. figure 6a, switch closed) applied to the input, the following conditions shall exist within the amplifier:

(a) The base voltage shall be no less than plus 2.85 volts nor greater than plus 3.85 volts.
(b) With the load removed, the output transistor leakage current shall not exceed 15 microamps.

3.4.6.2 False state - (output loaded).- With a false condition applied to the input (figure 6a, switch open). the following conditions shall exist within the amplifier:

(a) The base voltage shall be plus 2.6 plus or minus 0.4 volts.
(b) The collector voltage shall be plus 2.85 plus or minus 0.2 volts.

3.4.7 Input gates.- The diodes shall possess characteristics in consonance with their appropriate drawings. The forward and reverse characteristics shall fall within the test limit curves of room ambient operation.

3.4.8 Miscellaneous resistors.- The value of any resistor, not functionally connected to an active circuit, shall be in accord with the appropriate drawing.

3.4.9 Inhibit driver.

3.4.9.1 True input - (output loaded).- With the input and load circuits of figure 7 supplying a true signal input, the following conditions shall exist within each of the drivers:

(a) The voltage at the base of the transistor shall be plus 2.7 $\pm$ 0.3 volts.
(b) The output voltage shall be plus 2.85 plus or minus 0.5 volts.

3.4.9.2 False input - (output loaded).- With the load of figure 7 and a false input supplied to the input, the following conditions shall exist:

(a) The voltage at the base of the transistor shall be plus 5.5 $\pm$ 1 volt with respect to + 3 volts.
(b) The output voltage shall be minus 5.6 plus or minus 0.5 volts.

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3.5 Electrical performance for temperature extremes.- The following requirements shall apply for the conditions of 3.3.3.1 and 3.3.4.1 (as applicable).

3.5.1 Error indicator.

(a) With a true input to the indicator, and the load of figure 1 at the output, the neon bulb load shall flash at a visible rate.

(b) With a false input to the indicator, and the load at the output, the bulb shall not be lighted.

3.5.2 Minus 18 volt driver.- With the load of figure 2 connected for all tests, the following conditions shall exist:

(a) At minus 40°F

(1) A true input shall effect an output voltage no greater than plus 0.5.

(2) A false input shall effect an output voltage of minus 18.7 plus or minus 1.0 volts.

(b) At plus 135°F

(1) A true input shall yield a voltage no greater than plus 1 volt at the output.

(2) A false input shall yield a voltage no less negative than minus 15.2 volts at the output.

3.5.3 Oscillator (10KC).

(a) The oscillator output shall meet the requirements of figure 3.
(b) The diode (CR16), when grounded, shall cut off the output signal.

3.5.4 Phase detecting network.

(a) For the input of figure 4a, the voltage developed across the diode rectifier shall be no less than 110 volts D.C.
(b) For the input of figure 4b, the voltage developed across the diode rectifier shall be no greater than 15 volts D.C.

3.5.5 Regular driver.

(a) The input gate and load conditions of 3.4.5.1 shall be performed, and with a true input, the output transistor collector voltage shall be minus 6.2 plus or minus 0.4 volts.
(b) With a false signal applied, the output transistor collector voltage shall be no less negative than minus 0.1 volt and no more negative than minus 0.5 volt.

3.5.6 Write switch conversion amplifier.- With the load of figure 6b at the output of the amplifier, and the input network of figure 6a, the following conditions shall apply.

(a) With a true input, the collector voltage shall be minus 6 plus or minus 0.4 volts.
(b) With a false input, the collector voltage shall be plus 2.85 plus or minus 0.4 volts.

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3.5.7 Input gates.

3.5.7.1 AND and OR gates.- The diodes shall have a forward voltage drop of no more than 0.6 volt at 0.26 ma, nor a reverse leakage in excess of 25 microamps at a 15 volt bias.

3.5.8 Inhibit driver.- With the load of figure 7 connected for all tests, the following conditions shall exist:

(a) At minus 40°F.
(1) A true input shall effect a plus 2.85 plus or minus 0.6 volts at the output.
(2) A false input shall effect a minus 5.8 plus or minus 0.5 volts.
(b) At plus 135°F
(1) A true input shall effect a plus 2.7 plus or minus 0.5 volts at the output.
(2) A false input shall yield between minus 5.0 and minus 6.1 volts.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 Inspection provisions.

4.2.1 Submission of product.- Unless otherwise specified herein, or by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" as specified in MIL-STD-105.

4.2.2 Examinations and tests.- Except where otherwise specified herein, components and sub-assemblies shall be inspected in accordance with the "Inspection provisions for components and sub-assemblies" and the Sampling Plan Tables contained in MIL-F-13926. Examinations and tests related to Section 3 herein shall be performed in accordance with Tables I, II, and III and the test methods and procedures in 4.5. Examinations and tests for packaging, packing and marking shall be in accordance with MIL-P-14232 and Section 5 herein.

4.2.3 Acceptance and rejection.- Rejected lots shall be screened for defective characteristics. Removal or correction of defective [illegible] mittance of rejected lots shall be in accordance with "Acce-

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4.2.4 Functional tests.- The requirements and tests in Table I shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (60° to 90°F).

TABLE I
(100 Percent Inspection)

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Error indicator - true state	3.4.1.1	4.5.2.1.1
102.	Error indicator - false state	3.4.1.2	4.5.2.1.2
103.	Minus 18 volt driver - true input	3.4.2.1	4.5.2.2.1
104.	Minus 18 volt driver - false input	3.4.2.2	4.5.2.2.2
105.	10 KC oscillator	3.4.3	4.5.2.3.1
106.	Phase detecting network - phase 1	3.4.4.1	4.5.2.4.1
107.	Phase detecting network - phase 2	3.4.4.2	4.5.2.4.2
108.	Regular driver - true state	3.4.5.1	4.5.2.5.2
109.	Regular driver - false state	3.4.5.2	4.5.2.5.3
110.	Write switch conversion - true state	3.4.6.1	4.5.2.6.3
111.	Write switch conversion - false state	3.4.6.2	4.5.2.6.4
112.	Input gates	3.4.7	4.5.2.7
113.	Miscellaneous resistors	3.4.8	4.5.2.8
114.	Inhibit driver - true input	3.4.9.1	4.5.2.9.1
115.	Inhibit driver - false input	3.4.9.2	4.5.2.9.2

4.3 Control tests.

4.3.1 General.- One network E shall be selected at random as a control sample from each 100 produced. The control sample shall meet the requirements and tests in Table II.

TABLE II

	CONTROL TEST	REQUIREMENT	TEST PROCEDURES
301.	Fabrication	3.1	Applicable drawings - visual
302.	General specifications	3.2	MIL-F-13926 and MIL-C-46357 - visual

4.3.2 Environmental.- Three units shall be selected at random as a control sample from each 50 produced or from each month's production, whichever occ[illegible]irst. The control samples shall have met the requirements and tests in [illegible]hall then meet the requirements and tests in Table III.

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TABLE III

	CONTROL TEST	REQUIREMENT	TEST PROCEDURES
303	Vibration	3.3.1	4.5.1.1
304	Shock	3.3.2	4.5.1.2
305	Humidity - operating	3.3.4.1	4.5.1.3.2
306	Humidity - Non-operating	3.3.4.2	4.5.1.3.1
307	Salt spray	3.3.6	4.5.1.4
308	Operating temperature	3.3.3.1, 3.5	4.5.1.5, 4.5.3 thru 4.5.3.8
309	Storage temperature	3.3.3.2	4.5.1.5

4.3.3 Control test failure.- Should any one item of a control test sampling fail to meet the specified test requirements, acceptance of the product will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (See 4.2.3).

4.4 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply equipment in accordance with the applicable requirements of MIL-I-45607. The Government reserves the right to use the test equipment for its own independent inspections to the extent that such use will not unduly interfere with the contractor's delivery schedule.

4.4.1 Contractor design.- The contractor shall provide inspection equipment compatible with the "Test Methods and Procedures" specified in 4.5 and with the "Component Inspection" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.4.1.1 Test equipment.- In conjunction with 4.4.1, the following test equipment shall be utilized in the performance of the applicable tests as specified in 4.5.2

NOTE: The dc power supplies listed shall be variable to within 20 percent of the nominal voltage value, and have a regulation no worse than 2 percent. Unless otherwise specified, the peak-to-peak ripple shall be less than 100 mv, output impedance no greater than 2 ohms from dc to 500 kc.

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These supplies shall deliver the voltages and tolerances as specified for room ambient tests. They shall be temperature compensated as shown below, and these values shall be used for all temperature extreme tests.

Nominal (Volts)	Minus 40°F	Plus 135°F
-3	minus 3.21 ± 0.06 volts	minus 2.79 ± 0.06 volts
+3	plus 3.21 ± 0.06 volts	plus 2.79 ± 0.06 volts
-6	minus 6.42 ± 0.12 volts	minus 5.58 ± 0.12 volts
+6	plus 6.42 ± 0.12 volts	plus 5.58 ± 0.12 volts
-1.5	minus 1.6 ± 0.03 volts	minus 1.4 ± 0.03 volts
-18	minus 19.26 ± 0.36 volts	minus 16.74 ± 0.36 volts
+15	plus 16.05 ± 0.3 volts	plus 13.95 ± 0.03 volts
-32	minus 34.2 ± 0.8 volts	minus 29.8 ± 0.8 volts

Item Number	Description
1	Power Supply - Nominal value of plus 150 volts ± 2.32 volts current requirements of at least 200 ma. The peak-to-peak ripple and noise shall be less than 400 mv and the output impedance shall be less than 10 ohms - from dc to 500 kc.
2	Power Supply - Nominal value of minus 32 volts ± 1 volt, current requirement of 75 ma.
3	Power Supply - Nominal value of plus 3 volts ± 0.12 volts current rating of at least 30 ma.
4	Power Supply - Nominal value of minus 6 volts ± 0.18 volts current of at least 150 ma.
5	Power Supply - Nominal value of plus 6 volts ± 0.12 volts current rating of at least 100 ma.
6	Power Supply - Nominal value of minus 1.5 volts ± 0.45 volt current rating of at least 30 ma.
7	Power Supply - Nominal value of minus 18 volts ± 0.54 volt current rating of at least 100 ma.
8	Power Supply - Nominal value of plus 15 volts ± 0.45 volt current rating of at least 300 ma.
9	Power Supply - 3 phase, 120 volts rms plus or minus 2 percent line-to-neutral, 400 cps ± 2 cps.
10	Voltmeter, DC Digital - A range of at least .001 to 999 volts dc, or three significant figures. Accuracy of at least ± 0.2 percent of reading ± 1 count. Input impedance of at least 11 megohms on all ranges.

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Item Number	Description
11	Voltmeter, AC - Capable of measuring at least 225 volts rms. Accurate to within 3 percent of the scale reading. Input impedance of at least 10 megohms. Frequency response of at least 10c to 100 KC.
12	Oscilloscope Dual Channel - Rise time of scope plus measuring probe 50 nano seconds or less, sweep 50 nono seconds per centimeter or greater, input resistance greater than 8 megohms, input capacitance less than 25 pico-farads.
13	Ohmmeter - Capable of measuring dc resistance values to within 2 percent of the scale reading.
14	Miscellanious resistors, diode, neon light, switch, relay and capacitor.

Resistors

30 ohms, 1 watt, 1%
12000 ohms, 1/4 watt, 5%
470 ohms, 1 watt, 5%
68000 ohms, 1/4 watt, 5%
1000 ohms, 1/4 watt, 5%
2200 ohm, 10 watt, 5%
80 ohm, 1 watt, 1%

Neon Light - See Dwg. B8623458
Switch - Single pole, single throw (See Figures 1 thru 6)
Relay 10526121-3
Diode - IN99
Capacitor - 13 uf, 150 VDC, 10%

4.5 Test Methods and procedures.

4.5.1 Environmental.

4.5.1.1 Vibration.- The vibration test shall be made on a device with means provided for measuring and controlling the direction of vibration and adjusting the amplitudes and frequencies to keep them within the specified limits of 3.3.1. The measuring instruments of the testing device shall be capable of measuring the vibration amplitude within plus or minus 5 percent, and capable of measuring the frequency within plus or minus 2 percent. The test shall be performed in accordance with the requirements specified in 3.3.1 with the circuit board rigidly mounted in its normal operating position. Subsequent to the vibration test, the unit shall show no damage or defect resulting from the test when examined by visual and tactile means. The unit shall then be inspected as specified in 4.5.1.2 prior to determining compliance with requirements specified under 3.4.

4.5.1.2 Shock.- This test shall be conducted in accordance with Procedure II of the shock test defined in MIL-F-13926 except that the circuit board shall be shocked in the vertical plane only. The impact (shocks) and "G" value shall be in accordance with 3.3.2. A receiving device, representing the connector receptacle that receives the circuit board in the computer, shall be secured to the horizontal surface of the shock tester. Each end of the receiving device shall be clamped such that the female receiving slot will hold the unit in a vertical plane. The unit will then be inserted into the receiving device and shocked. Subsequent to imposing the number of impacts required, a visual examination shall be made for damages or defects to determine compliance with 3.3.2 and then the unit will be inspected as specified in 4.5.1.3 prior to determining compliance with 3.4 as required by 3.3.2.

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4.5.1.3 Humidity.

4.5.1.3.1 Non-operating.- This test shall be conducted in accordance with the conditions in 3.3.4.2 and the humidity test defined in Procedure I of MIL-F-13926 except that the 4-1/2 hour period shall be reduced to 2 hours, the 6 hour period reduced to 2 hours, and the total test period shall not exceed 5.5 hours per cycle. The unit shall then be thoroughly dried and then subjected to the test in 4.5.1.3.2 prior to meeting the specified requirements of 3.4 as required by 3.3.4.2.

4.5.1.3.2 Operating.- This test shall be conducted in accordance with the conditions in 3.3.4.1 and the humidity test defined in Procedure III of MIL-F-13926 except that the humidity shall be maintained at 97 $\pm$ 5 percent for the 24 hour period and the test shall cease at the end of that period (Method 106 of MIL-STD-202 shall be excluded). Throughout the test the unit shall be operated in accordance with 4.5.3 to determine compliance with 3.3.4.1.

4.5.1.4 Salt spray.- The circuit board shall be subjected to the conditions in 3.3.6 and the salt spray test defined in MIL-STD-202, Condition B. Immediately after exposure, the unit shall be inspected for any physical damage, then tested in accordance with 4.5.2 for compliance with 3.3.6.

4.5.1.5 Temperature (Operating and storage).- After being subjected to all environmental tests specified above, the circuit board shall be subjected to Test Procedure I specified in MIL-F-13926, except the temperatures specified in 3.3.3 shall apply. While at the operating temperatures of 3.3.3.1, the unit shall be tested in accordance with 4.5.3 to determine compliance with the requirements of 3.5. as required by 3.3.3.1. The temperature shall then be returned to standard temperature (60° to 90°F) and the unit shall be tested in accordance with 4.5.2 to determine compliance with 3.3.3.2.

4.5.1.6 Fungus.- To determine compliance with 3.3.5, one of the first 5 network E boards produced shall be submitted for testing in accordance with the fungus test defined in MIL-F-13926. After approval, subsequent acceptance for fungus requirements may be by certification.

4.5.2 Performance, Electrical.

4.5.2.1 Error indicator test.- Compliance with 3.4.1 shall be determined by the following test: Connect a minus 3 volt supply (item 2, 4.4.1.1) to pin 22B. Connect plus 150 volt supply (item 1) to pin 10B.

4.5.2.1.1 True state - static condition (output loaded).

(a) Connect the load of figure 1 across pins 7B and 11B.
(b) Apply minus 6 volts (item 4) to pin 5A. The neon bulb shall flash at a visible rate.

4.5.2.1.2 False state - static condition (output loaded).

(a) Remove the minus 6 volt input and ground pin 5A. The voltage at TP1 shall be between minus 2.1 and minus 3.0 volts.
(b) The neon bulb shall not light.

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4.5.2.2 Minus 18 volt driver tests.- Compliance with 3.4.2 shall be determined by the following tests. Connect the load of figure 2 to pin 64A. Make the following power connections:

-18V	to	47B
-32V	to	61A
+ 6V	to	64B
+ 3V	to	53B
+15V	to	20B
ground		62B

4.5.2.2.1 True state.- Connect the input circuit of figure 2 to pin 61B.

(a) The voltage at TP26 shall be plus 2.8 plus or minus 0.3 volts.
(b) The voltage at TP27 shall be plus 3 plus or minus 0.5 volts.
(c) The voltage at pin 64A shall be at ground plus or minus 0.5 volt.

4.5.2.2.2 False state - Ground pin 61B.

(a) The voltage at TP26 shall be plus 5.5 $\pm$ 1 volt.
(b) The voltage at TP27 shall be minus 18 $\pm$ 0.8 volts.
(c) The voltage at pin 64A shall be between minus 16.8 and minus 18.5 volts.

4.5.2.3 10KC Oscillator tests.- Compliance with 3.4.3 shall be determined by the following test. Connect minus 6 volts to pin 3A, plus 6 volts to pin 4A, and bring the ground returns of the power supplies to pin 1A.

4.5.2.3.1 Waveshape.

(a) Connect an oscilloscope between pins 6A and ground (pin 1A). The observed waveform shall be as directed in figure 3.
(b) When pin 5B is grounded, the wave observed in (a) shall be attenuated to zero.

4.5.2.4 Phase Detecting Network.- Compliance with 3.4.4 shall be determined by the following test. Connect the loads of Figure 4d to the appropriate pins.

4.5.2.4.1 Phase one operation.-

(a) Apply 3 phase, 120 volt phase to neutral, 400 cps (item 9, 4.8.1) as per figure 4a. Momentarily set switch 1 (see figure 4d).
(b) Connect an ac voltmeter (item 11, 4.8.1) between TP23 and Pin 35A. The voltage shall be no greater than 30 volts rms.
(c) Measure the ac voltage between pins 32A and 35A. It shall be no less than 155 volts rms.
(d) The dc voltmeter (item 10) shall be connected between pins 36A and 39B. The voltage shall be no less than 130 VDC.

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4.5.2.4.2 Phase two operation.

(a) Switch the inputs made in 4.9.2.4.1 (a) to conform to figure 4b. Momentarily depress switch 1 (figure 4d).
(b) Connect an ac voltmeter across TP23 and pin 35A. The voltage shall be no less than 155 volts rms.
(c) Measure the ac voltage between pins 32A and 35A. It shall be no greater than 30 volts rms.
(d) Connect a dc voltmeter between pins 40B and 39B. The voltage shall not exceed 15 VDC.

4.5.2.5 Regular driver test.- Compliance to 3.4.5 shall be determined by the following tests. Prior to energizing the module, the input gate of figure 5a shall be connected to pin 7A, and the load of figure 5b shall be connected to pin 12B.

4.5.2.5.1 Power supplies.- The following power connections shall be made:

Minus 1.5 volts at 14B
Minus 18 volts at 6B
Minus 6 volts at 23B
Plus 15 volts at 20B

The dc returns of each of the power supplies shall be connected to pin 21B.

4.5.2.5.2 True state - static condition (output loaded).

(a) Connect minus 6 plus or minus 0.18 volts to the input gate.
(b) Measure the voltage at TP19; it shall be minus 1.6 plus or minus 0.1 volts.
(c) The voltage measured at TP21 shall be no less than 0.1 volt nor any greater than 0.5 volt.
(d) With the load removed the voltage measured at pin 12B be between 0 and + 0.2 volts with respect to the -6 volts.

4.5.2.5.3 False state - static condition - (output loaded).

(a) Replace the minus 6 volts at the input gage with power ground.
(b) Measure the voltage at TP21. It shall be minus 0.52 plus or minus 0.23 volt.
(c) The voltage measured from TP19 to TP21 shall be minus 6.2 plus or minus 0.31 volts.
(d) The measured voltage at pin 12B shall be between 0 and -.5 volts.

4.5.2.6 Write switch conversion amplifier tests.- Compliance to 3.4.6 shall be determined by the following tests.

4.5.2.6.1 Power and ground connections.

(a) Connect the minus 6 volts to pins 8B and 44B. Connect the plus 3 volts to pins 16B and 53B.
(b) The dc returns of the power supplies shall be connected to pins 8A and 44A.

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4.5.2.6.2 Input and load connections.

(a) Connect the input network (see figure 6a) to eachof the pins of column A, Table I.
(b) Connect loads (as per figure 6b) to each of the pins of column C, Table I.

4.5.2.6.3 True state (output loaded).

(a) Close the input switch of figure 6a.
(b) The voltage at each of the points of column B, Table IV, with respect to ground, shall be no less than plus 2.85 volts, but not in excess of plus 3.85 volts.
(c) With the load removed, the voltage at each of the pins of column C, Table V, shall be between 0 and +0.33 volts with respect to the minus 6 volt power level.

4.5.2.6.4 False state - (output loaded).

(a) Open the input switch of figure 6a.
(b) The voltages measured at the points of column B, Table IV shall be plus 2.6 plus or minus 0.4 volts.
(c) The voltages at each of the pins of column C, Table IV shall be plus 2.85 plus or minus 0.2 volts.

TABLE IV

INPUT	BASE	OUTPUT
Column A	Column B	Column C
22A	TP7	14A
20A	TP9	12A
18A	TP11	10A
58A	TP13	50A
56A	TP15	48A
54A	TP17	46A
53A	TP18	45A
55A	TP16	47A
57A	TP14	49A
17A	TP12	9A
19A	TP10	11A
21A	TP 8	13A

4.5.2.7 Input gate tests.- Compliance to 3.4.7 shall be determined by the following tests.

4.5.2.7.1 AND gate.

(a) Connect minus 18 volts (item 7) to pin 37A. Ground each of pins 38A, 39A, 40A, 41A, 42A, and 43A, one-at-a-time. As a pin is grounded, measure the voltage at pin 43B; it shall not exceed minus 0.3 volts. After a measurement, remove the ground connection and ground the next pin.

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(b) Replace the minus 18 volts at pin 37A with a plus 15 volts. Repeat the procedure in (a) above except the voltage measured at pin 43B shall be no less than 10.5 volts with respect to ground +, 15 volts.

4.5.2.7.2 Diode Gate.

(a) Ground pins 30A, 31A and 30B.
(b) Apply minus 18 volts through a 12,000 ohm resistor to each of the pins specified in column A, Table V, one at a time. The voltage measured at each of the pins in column A, Table V, shall be less than minus 0.7 volts with respect to ground.
(c) Apply plus 15 volts through a 0.5 megohm resistor to each of the pins specified in column A, Table V, one at a time. The voltage measured at each of the pins in column A, Table V, shall be less than plus .05 volts with respect to +15 volts.

TABLE V

A	B
29A)	
29B)	
28A)	
28B)	30A
27A)	
27B)	
26A)	
26B)	
25A)	
25B)	31A
24A)	
24B)	30B

4.5.2.8 Resistor tests.- Compliance to 3.4.8 shall be determined by connecting an ohmmeter between pins 44A and 37B. The resistance measured shall be 2000 ohms ± 1 percent. The resistance between pins 42B and 37B shall be 4000 ohms plus or minus 1 percent. The resistance measured between 60A and 65A shall be 120 ohms plus or minus 5 percent.

4.5.2.9 Inhibit driver tests.- To determine compliance with 3.4.9, connect the load of figure 7 to each of pins 55B and 57B. Connect plus 3 volts to pin 53B, minus 6 volts to 44B, minus 18 volts to 47B, and plus 15 volts to 20B.

4.5.2.9.1 True state.- Connect the input circuit of figure 7 to each of pins 45B and 46B.

(a) The voltage at TP24 and TP25 shall be plus 2.7 ± 0.3 volts.
(b) The voltage at pins 55B and 57B shall be plus 2.85 plus or minus 0.5 volts.

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4.5.2.9.2 False state.- Apply plus 3 volts to pins 45B and 46B.

(a) The voltage at TP24 and TP25 shall be plus 5.5 ± 1 volt with respect to +3 volts.
(b) The voltage at pins 55B and 57B shall be minus 6 plus or minus 0.5 volts.

4.5.3 Temperature extremes.- The following tests shall be performed at the test condition of 4.5.1.3.2 (as spplicable), and 4.5.1.5 to determine compliance with 3.5.

4.5.3.1 Error indicator.- The following test shall determine compliance with 3.5.1.

(a) Connect the load of figure 1 across pins 7B and 11B. Connect minus 3 volts to 22B, and plus 150 volts to 10B as per 4.4.1.1.
(b) Apply minus 6 volts to 5A. The neon bulb shall flash at a visible rate.
(c) Remove the minus 6 volts and ground pin 5A. The neon bulb shall not light.

4.5.3.2 Minus 18 volt driver.- Make all the load and power connection of 4.5.2.2. Compliance with 3.5.2 shall be determined by the following test:

(a) At minus 40°F.

(1) Connect the input circuit of figure 2 to pin 61B (use proper voltage as per 4.4.1.1,) and measure a voltage no greater than plus 0.5 volt at 64A.

(2) Ground pin 61B, and measure minus 18.7 plus or minus 1.0 volts at 64A.

(b) At plus 135°F.

(1) Connect the input of figure 2 to pin 61B, and measure a voltage no greater than plus 1 volt at 64A.

(2) Ground 61B, and measure a voltage no less negative than minus 15.2 volts at 64A.

4.5.3.3 10KC Oscillator.- Compliance with 3.5.3 shall be determined by performing all the operations of 4.5.2.3 using the supply voltage as per 4.4.1.1.

4.5.3.4 Phase detecting network.- Compliance with 3.5.4 shall be determined by the following tests:

(a) Apply 3 phase, 120 volts, 400 cps to pins 34B, 33A and 36B in the phase relation of figure 4a. The dc voltage measured from pin 39B to pin 36B shall be no less than 110 volts DC.
(b) Reverse the phases as shown in figure 4b. The voltage measured from pin 39B to pin 36B shall be no greater than 15 volts DC.

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4.5.3.5 Regular driver.- Compliance with 3.5.5 shall be determined by the following tests:

(a) Make the power connections of 4.5.2.5.1 and the input gate and load connections of 4.5.2.5. (ref. 4.4.1.1)
(b) With a true signal at the input gate (minus 6 plus or minus 0.18 volts) the voltage at pin 12B shall be minus 6.2 plus or minus 0.4 volts.
(c) With a false signal to the input gate (ground the input to the input gate) the voltage at pin 12B shall be no less negative than minus 0.1 volt, nor any more negative than minus 0.5 volt.

4.5.3.6 Write switch conversion amplifier.- Compliance with 3.5.6 shall be determined by the following tests:

(a) With the condition and external connections of 4.5.2.6.1 (ref. 4.4. 4.4.1.1) and 4.5.2.6.2, and a true input, the voltages at the pins of column C, Table I, shall be minus 6 plus or minus 0.4 volts.
(b) With a false input, the voltages at the pins of column C, Table I, shall be plus 2.85 plus or minus 0.4 volts.

4.5.3.7 Input gates.- (use the appropriate voltages of 4.4.1.1) to determine compliance with 3.5.7.

4.5.3.7.1 AND gates.- Perform 4.5.2.7.1 except that in (a) the measured voltage shall not exceed 0.6 volt, and in (b) the measured voltage shall be no less than 13.25 volts.

4.5.3.8 Inhibit driver.- Make all load and power connections of 4.5.2.9, and perform the following tests to determine compliance with 3.5.8.

(a) at minus 40°F.

(1) Connect the input circuit of figure 7 to each of pins 45B and 46B. The voltage at pins 55B and 57B shall be plus 2.85 plus or minus 0.6 volts.

(2) Apply plus 3 volts to pins 45B and 46B. The voltage at pins 55B and 57B shall be minus 5.8 plus or minus 0.5 volts.

(b) at plus 135°F.

(1) Connect the input circuit of figure 7 to pins 45B and 46B. The voltage at 55B and 57B shall be plus 2.7 plus or minus 0.5 volts at the output.

(2) Apply plus 3 volts to 45B and 46B. The voltage at pins 55B and 57B shall be between -5.0 and -6.1 volts.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing.- Preservation, packaging and packing shall be in accordance with Master Packaging Data Sheet M14002.

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6 NOTES

6.1 Intended use.- The nine functional circuits contained on Network E, with their intended use, are as follows:

6.1.1 Error indicator.- Indicates any error in FALT true-false logic of the flip-flops, primary gates, or drivers.

6.1.2 Minus 18 volt driver.- Supplies the minus 18 volts to the flip-flops to test the FIT and FOT lines.

6.1.3 Oscillator, 10KC.- Triggers a one-shot multivibrator to produce the clock pulse. It is the primary oscillator which triggers the clock circuit.

6.1.4 Phase detecting network.- A phasing error in the application of the 3-phase input to the FALT system would cause the cooling fans for the unit to run in reverse. The phase detecting network compensates for any error and corrects the voltage application to the fans.

6.1.5 Regular driver.- The regular driver provides power amplification for driving the logic gates.

6.1.6 Write switch conversion amplifier.- The write switch conversion amplifier provides current amplification to crosspoints and for matching of write switches.

6.1.7 Input gates.- The input gates act as OR logic gates for minus 6 volts and AND gates for 0 volts. This condition produces a zero for any false logic. They are utilized in the confidence tests to insure line continuity.

6.1.8 Inhibit driver.- Inhibits the one clock or zero clock pulses for logic action.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable level of preservation, packaging, and packing.
(c) Applicable stock number.
(d) Applicable packaging data number.
(e) A block of serial numbers to identify each circuit board.

6.3 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

Custodian:

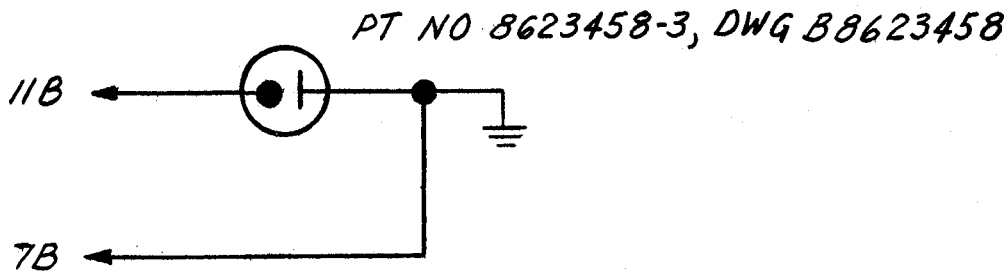
Army - MU

Preparing activity:

Army - MU

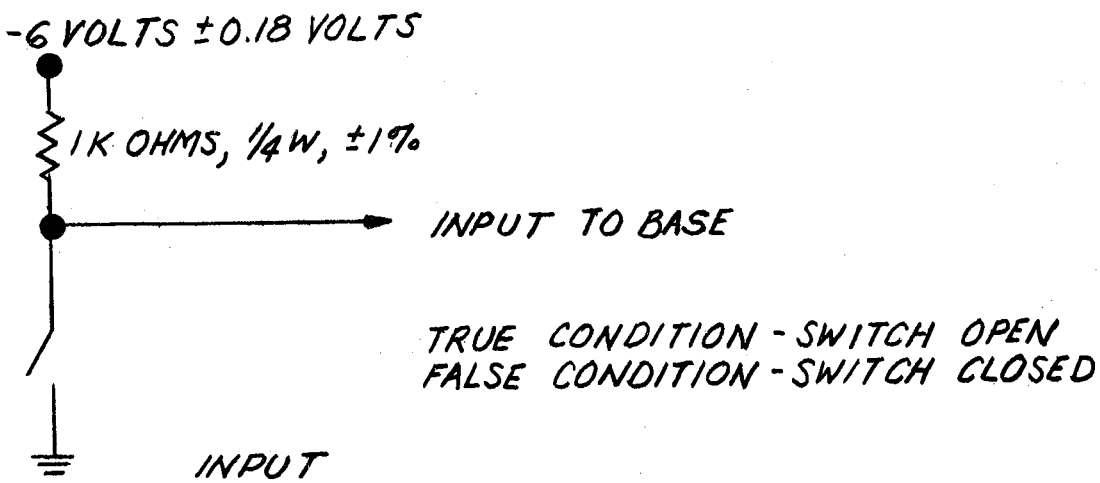
Project No. 1220-A182Y

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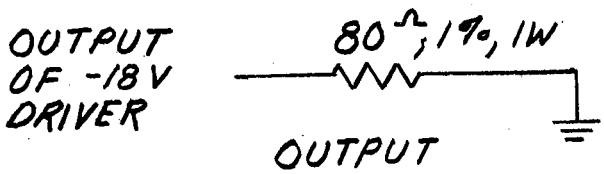


LOAD FOR ERROR INDICATOR

FIGURE 1



(a)

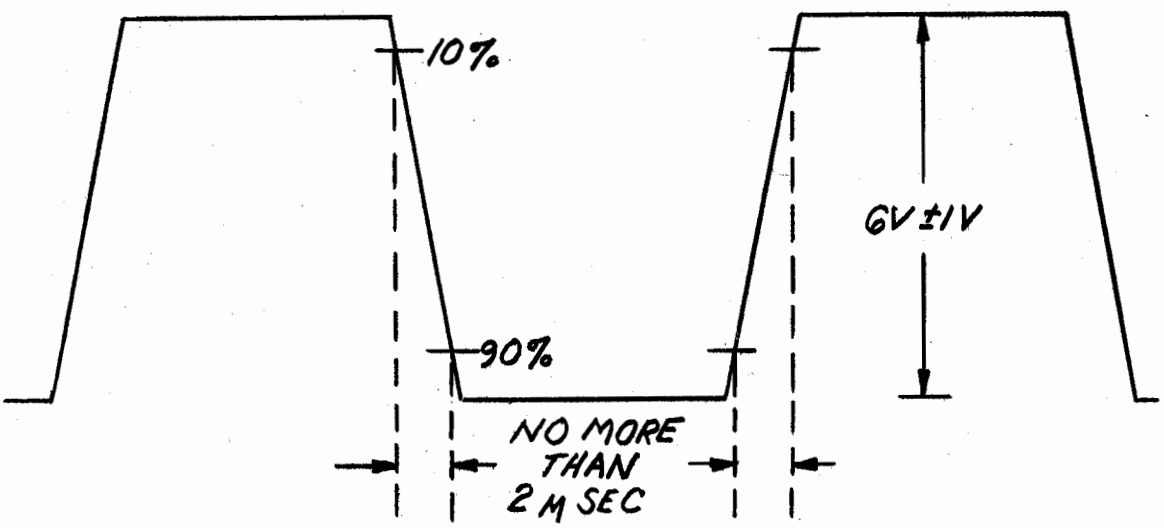


(b)

MINUS 18 VOLT DRIVER

FIGURE 2

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f @ ROOM AMBIENT	10 KC ± 1 KC
f @ +125°F	NO LESS THAN 8.5 KC
f @ -40°	NO GREATER THAN 12.5 KC

10 KC OSCILLATOR WAVESHAPE

FIGURE 3

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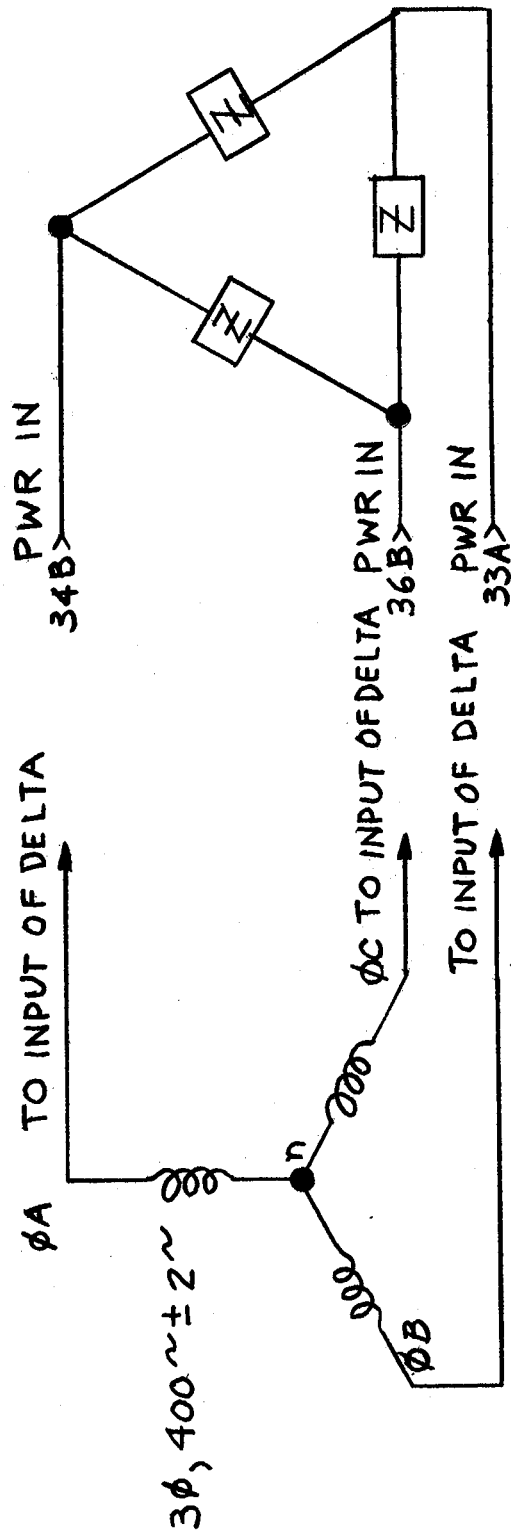


FIGURE 4A

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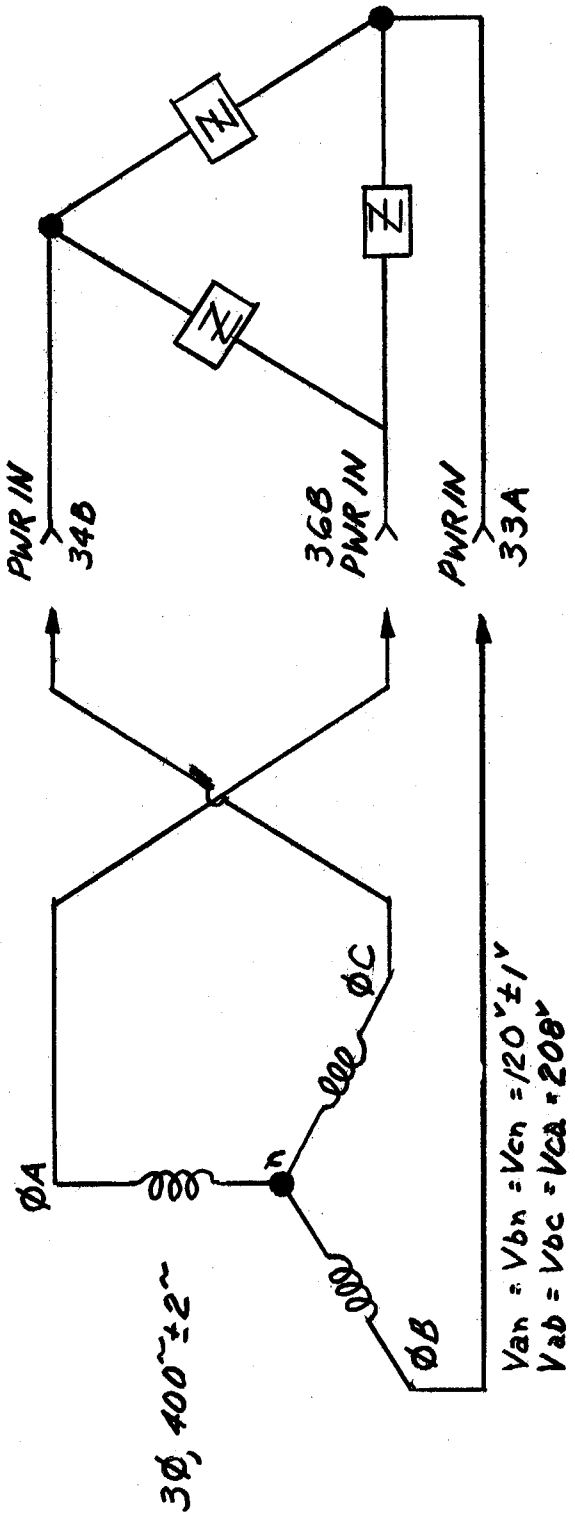


FIGURE 4b

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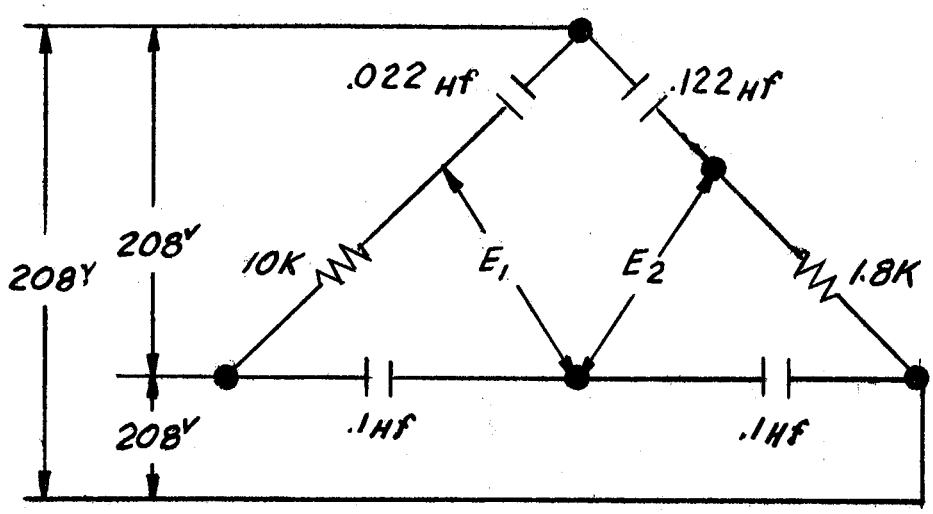
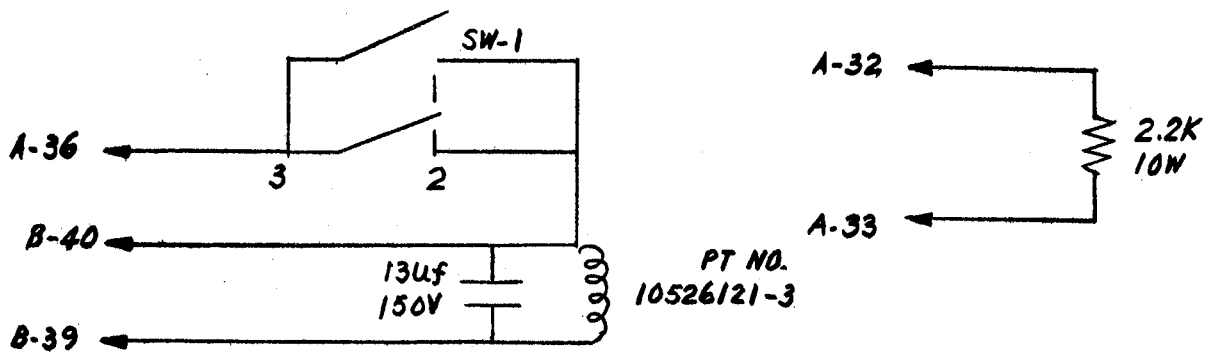
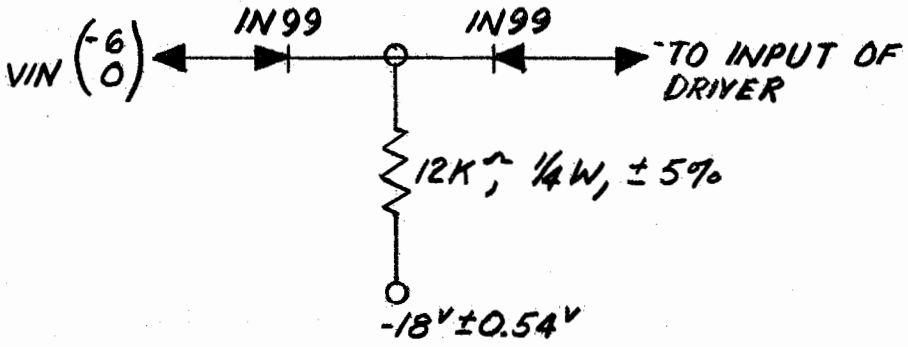


FIGURE 4C



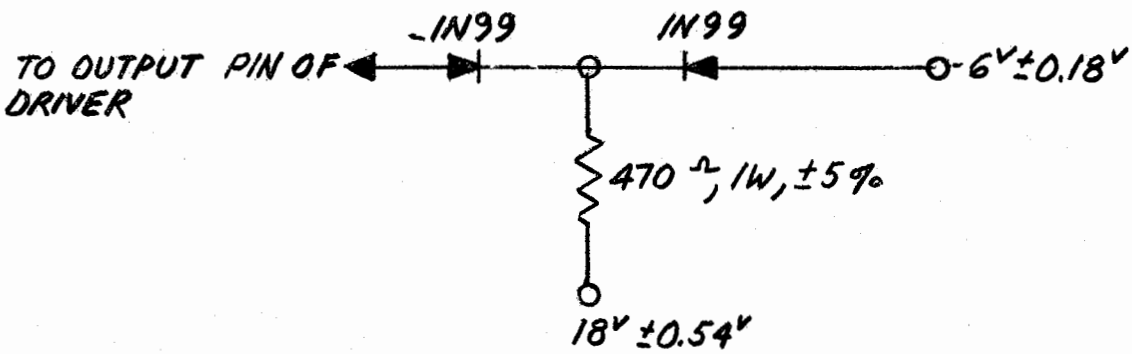
LOADS FOR PHASE SENSING NETWORK
FIGURE 4D

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INPUT GATE FOR REGULAR DRIVER

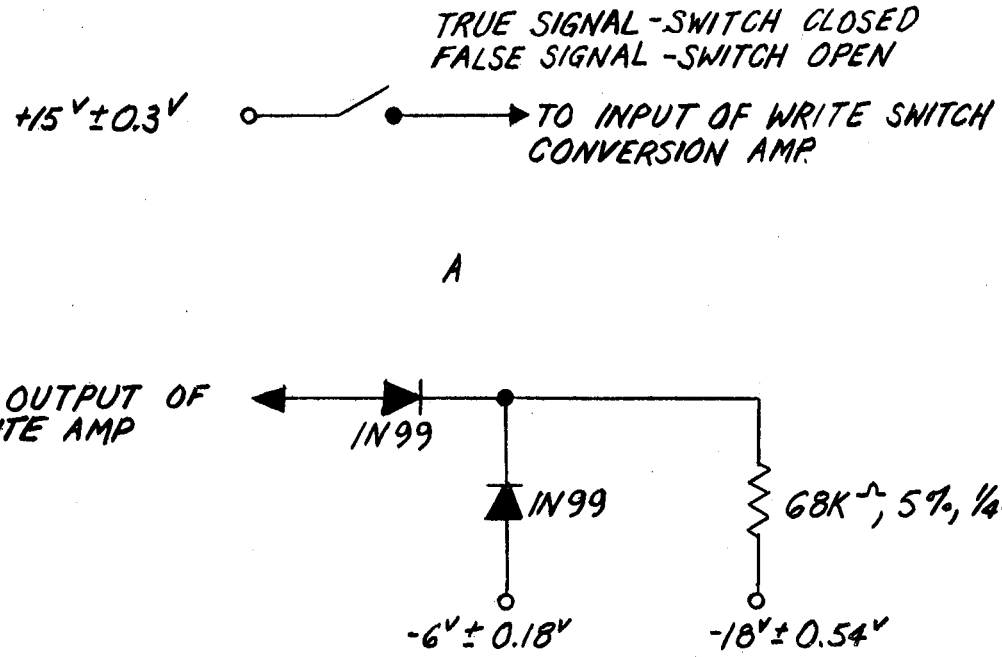
FIGURE 5a



OUTPUT LOAD FOR REGULAR DRIVER

FIGURE 5b

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INPUT & OUTPUT NETWORKS FOR WRITE SWITCH CONVERSION AMPLIFIER

FIGURE 6

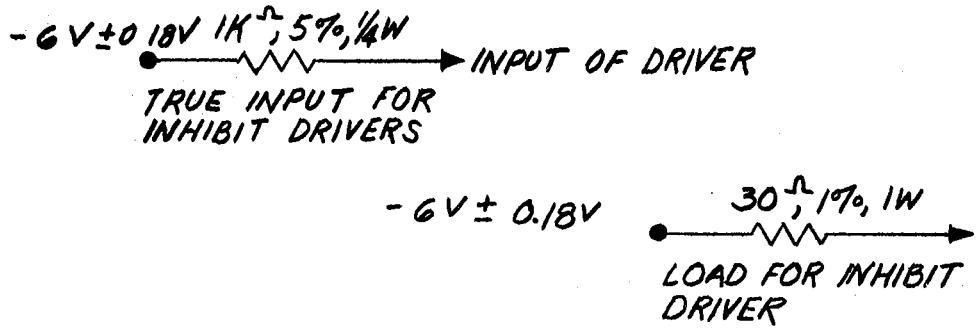


FIGURE 7

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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